Phase B4 Synthesis — The Complete Composition Pairs Architecture of Paper 2

How 28 Individual Composition Pairs and Clusters Form Seven Higher-Order Architectural Clusters, Constitute the Governance Interdependency Map of the CKS Three-Level Instinct/Reasoning-Separated Architecture, and Close Phase B4

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Attribution

This work derives from and formalizes the instinct/reasoning separation pattern introduced in "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026), the second paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026). It does not introduce new axioms. Its sole contribution is to synthesize the 28 individual composition pairs and clusters documented in derivation notes B4.02 through B4.29 into seven higher-order architectural clusters, and to demonstrate that these 28 pairs together constitute the complete governance interdependency map of Paper 2's three-level instinct/reasoning-separated architecture.

Abstract

Phase B4 of the CKS derivation note series (notes B4.01–B4.30) documents 28 composition pairs and clusters — pairwise and multi-concept interdependencies within Paper 2's architectural commitments, and between those commitments and relevant commitments from Paper 1. This synthesis note, B4.30, completes Phase B4 by organizing the 28 pairs into seven higher-order architectural clusters: (1) the instinct/reasoning substrate cluster, (2) the three-level structure cluster, (3) the lifecycle governance cluster, (4) the evolution architecture cluster, (5) the substrate and record cluster, (6) the domain and composition cluster, and (7) the recursive governance authority cluster. The note shows how cross-cluster pairs tie the clusters together, characterizes the complete set of 28 pairs as formalized prior art for Paper 2's governance interdependency architecture, and closes Phase B4 with a preview of Phase B5 (operational tests).

1. Why Phase B4 synthesis matters — completing the composition pairs map

Composition pairs are not miscellaneous derivations. Each pair formalizes a structural interdependency: an architectural reason why two or more of Paper 2's commitments require each other to function as the paper describes. A single commitment stated in isolation can be understood; two commitments whose interaction is named and formalized constitute a constraint — a piece of architectural prior art that cannot be later claimed as a novel discovery by a third party, because the interdependency has now been publicly named, dated, and attributed.

Phase B4 opened with a framing note (B4.01) establishing the composition pair method: that the governance architecture of Paper 2 is not fully characterized by any single commitment, but by the network of interdependencies among those commitments. Notes B4.02 through B4.29 then worked through 28 individual pairs and clusters, each establishing one interdependency with precision and operational derivation.

Twenty-eight pairs is a large number. The synthesis question is not "did we cover everything?" but "what does the complete set reveal that no individual pair shows?" The answer is the interdependency map: a structured view of how Paper 2's architectural commitments are organized into functional sub-systems, how those sub-systems depend on each other, and which commitments occupy load-bearing positions in more than one sub-system. That map is what this note produces.

The map has two defensive purposes. First, it makes the individual pairs easier to navigate — a reader seeking to understand any one interdependency can locate it within its functional cluster and understand its neighbors. Second, and more important for the defensive publication purpose, it formalizes the complete governance interdependency architecture as prior art in one place. The map covers the entire design space; there is no gap in which an adjacent formalization could claim novelty.

2. Seven higher-order clusters and their roles

The 28 pairs organize into seven clusters, each covering a distinct functional sub-system of Paper 2's architecture. The clusters are not arbitrary groupings — each is defined by a shared architectural question the member pairs answer together.

Cluster 1 — Instinct/reasoning substrate cluster

Constituent pairs: B4.02 (B1.01 + B1.06), B4.03 (B1.01 + B1.07), B4.17 (B1.01 + A1.12), B4.18 (B1.13 + B1.01)

Paper 2's foundational move is the instinct/reasoning separation (B1.01): the LLM is the instinct layer (fast, pattern-matched, no-reasoning-needed) and the CKS substrate is the reasoning layer (deliberate, human-governed, corrective). Cluster 1 covers the four pairs that establish how this separation is implemented, extended, characterized, and maintained.

B4.02 establishes that the separation is implemented via the two-layer structure (B1.06 — DNA layer and action layer within every cell): the DNA layer encodes the governed, stable reasoning-side knowledge; the action layer accumulates lived experience on the instinct-side path. The two-layer distinction is not merely organizational; it is what makes the instinct/reasoning separation architecturally tractable rather than metaphorical.

B4.03 establishes that the separation extends across cell boundaries via the expression mechanism (B1.07): cells do not need to carry the full architecture in active form at all times; the expression substrate governs which portions of the DNA layer are active for a given goal. Expression is what makes the instinct/reasoning separation scale from the cell level to the aspect and Self levels without requiring uniform activation everywhere.

B4.17 establishes that the separation is characterized by the labor allocation framework (A1.12 from Paper 1): the three labor modes — direct human, LLM under rule, stable cell — correspond to different points on the instinct/reasoning spectrum, and the instinct/reasoning separation is what makes the three-mode framework coherent rather than arbitrary.

B4.18 establishes that the separation is maintained through evolutionary time via the mutation governance commitment (B1.13): mutations to the instinct layer (LLM weight changes) and mutations to the reasoning layer (substrate content changes) are governed differently because they are different kinds of change in the architecture, not just change of different scope.

Together, the four pairs constitute a complete architectural ecosystem for the instinct/reasoning separation: how it is physically instantiated (B4.02), how it extends across scope levels (B4.03), how it structures labor (B4.17), and how it is preserved under evolution (B4.18).

Cluster 2 — Three-level structure cluster

Constituent pairs: B4.04 (B1.02 + B1.17), B4.05 (B1.02 + B1.20), B4.20 (B1.03 + B1.04 + B1.05)

Paper 2's three-level structure (B1.02 — cell, aspect, Self) is not a simple hierarchy. Cluster 2 covers the three pairs that establish how the levels function: their relational determination, their recursive governance, and their co-requirement as a complete entity set.

B4.04 establishes that levels require functional determination via relational roles (B1.17): the same underlying CKS artifact can participate as a cell, as part of an aspect, or as a component of different aspects simultaneously, depending on what the structure is for. Levels are not intrinsic properties of artifacts; they are relational properties assigned by structural purpose.

B4.05 establishes that the three levels require recursive governance (B1.20): Paper 1's architectural commitments hold at every level without modification. Recursive governance is not governance at the top level cascading down; it is governance instantiated independently at each level with level-appropriate scope.

B4.20 establishes that the three entity types (cell, aspect, Self — B1.03, B1.04, B1.05) are co-required: none of the three can be removed without the architecture losing its capacity to represent multi-aspect, unified intelligence under governed evolution.

Cluster 3 — Lifecycle governance cluster

Constituent pairs: B4.06 (B1.09 + B1.10 + B1.11), B4.21 (B1.09 + B1.14), B4.22 (B1.11 + B1.14), B4.29 (B1.09 + B1.20)

Lifecycle events — birth (B1.09), mating (B1.10), death (B1.11) — are what give the three-level structure its temporal dimension. Cluster 3 covers the four pairs that establish how lifecycle events are governed: their core trinity, their relationship to directed selection, and their operation under recursive governance.

B4.06 establishes the lifecycle trinity as an integrated cluster: birth initiates a new cell, aspect, or Self under human governance; mating extends lineage by combining parental content under governed orchestration; death closes a lifecycle by obsolescence or lineage supersession. The three operations together cover the complete temporal arc of any CKS entity.

B4.21 establishes that birth is not independent of evolutionary dynamics: directed selection (B1.14) shapes which offspring are born and under what conditions. Birth is the point at which selection pressure from prior generations meets the architecture's capacity for governed origination.

B4.22 establishes the symmetric point for death: directed selection shapes which entities are retired and when. Lineage supersession in particular — where a superior offspring causes parent retirement — is a direct instantiation of directed selection operating through the death mechanism.

B4.29 establishes that birth under recursive governance (B1.20) is architecturally richer than birth under a single governance level: a new cell is born under cell-level governance; a new aspect under aspect-level governance; a new Self under Self-level governance. The level at which birth occurs determines which governance scope the new entity inherits.

Cluster 4 — Evolution architecture cluster

Constituent pairs: B4.07 (B1.12 + B1.13 + B1.14 + B1.15), B4.08 (B1.15 + B1.16 upward), B4.09 (B1.07 + B1.16 downward), B4.23 (B1.12 + B1.17), B4.27 (B1.15 + A1.10)

Evolution in Paper 2 operates through three mechanisms in productive tension: instinct evolution (B1.12), DNA evolution (B1.13), and action-feedback evolution (B1.15). Cluster 4 covers the five pairs that establish how evolution operates — its mechanisms, its directional channels, and its governance requirements.

B4.07 establishes the evolution cluster: the three mechanisms plus directed selection (B1.14) are co-required. Directed selection is what prevents evolutionary drift — without it, each mechanism can change the architecture in any direction; with it, changes are governed toward fitness criteria specified by humans at the appropriate level.

B4.08 establishes that evidence-based upward evolution (action-feedback B1.15 + bidirectional evolution B1.16) flows from lived experience toward DNA refinement: accumulated action-layer records provide the evidence base for proposing DNA-layer changes. Upward evolution is what makes the architecture learn from operation rather than only from explicit design.

B4.09 establishes the downward direction: expression (B1.07) enables expression-based downward evolution via the bidirectional channel (B1.16). When the DNA layer changes, the expression substrate propagates those changes to cell behavior without requiring each cell to be individually updated.

B4.23 establishes that evolution operates through role changes as well as content changes: evolutionary transitions can change an entity's relational roles (B1.17). Evolution is not only content mutation; it is structural reconfiguration.

B4.27 establishes the governance requirement for evidence-based evolution: the determinism contract (A1.10 from Paper 1) is what makes action-layer records reliable evidence. If substrate representations were non-deterministic, the same operational history could produce inconsistent records, making evidence-based evolution ungovernable.

Cluster 5 — Substrate and record cluster

Constituent pairs: B4.16 (A1.08 + A1.07), B4.12 (B1.14 + A1.07), B4.13 (B1.15 + B2.26), B4.19 (B1.06 + B1.14), B4.24 (B1.18 + B1.06)

The substrate is not a passive record. Cluster 5 covers the five pairs that establish substrate authority, the retraceability requirement, and the specific roles of the DNA layer and action layer in governing and enabling evolution.

B4.16 establishes the substrate authority and retraceability pair (A1.08 + A1.07): the substrate is the source of truth for all authoritative state, and every piece of authoritative state must carry the six-field provenance metadata that makes path retraceability possible. These two commitments are inseparable — a substrate that is authoritative but not retraceable cannot support governed evolution.

B4.12 establishes that directed selection (B1.14) requires path retraceability (A1.07): selection criteria must be traceable to the humans who specified them, and selection outcomes must be traceable to the evidence and criteria that produced them. Selection without retraceability is not governed selection; it is opaque change.

B4.13 establishes that action-feedback evolution (B1.15) requires the action layer architecture from B2.26: the action layer is where feedback accumulates, and the action layer's structure determines what kinds of evidence can be generated.

B4.19 establishes that the two-layer structure (B1.06) is what makes directed selection architecturally tractable: directed selection operates over the DNA layer and is informed by the action layer. Without the two-layer separation, directed selection has no target — it cannot distinguish between what defines the cell and what the cell has done.

B4.24 establishes that content-domain (B1.18) and the DNA layer are structurally coupled: the DNA layer of each cell in the content domain is what the higher level reads when it asks pattern questions. Content-domain is not just a membership set; it is a DNA-layer reading surface.

Cluster 6 — Domain and composition cluster

Constituent pairs: B4.10 (B1.18 + A1.13), B4.11 (B1.18 + B1.19), B4.25 (B1.04 + B1.08)

Composition in CKS is not aggregation. Cluster 6 covers the three pairs that establish how content-domain enables composition requirements, how it relates to cross-level access, and how modularity enables aspect coordination.

B4.10 establishes that content-domain (B1.18) activates Paper 1's composition requirements (A1.13): when an aspect or Self operates over a content domain spanning multiple substrates, all of Paper 1's composition requirements apply — per-substrate human governance preservation, conflict preservation across substrate boundaries, and the prohibition of composition that silently resolves conflicts that should be preserved.

B4.11 establishes that content-domain and cross-level access (B1.19) together define the access architecture: content-domain specifies what is in scope for a given level's pattern questions; cross-level access specifies the channels through which higher levels reach lower-level content.

B4.25 establishes that the aspect entity type (B1.04) and the modularity commitment (B1.08) are architecturally coupled: modularity from the cell up is what makes aspects possible as purpose-defined coordination arrangements rather than fixed organizational units.

Cluster 7 — Recursive governance authority cluster

Constituent pairs: B4.14 (B1.19 + B1.20), B4.15 (B1.20 + A2.47), B4.26 (A1.03 + B1.14), B4.28 (A1.01 + B1.20)

Governance authority in CKS is not a single rule applied uniformly. Cluster 7 covers the four pairs that establish the complete recursive governance authority architecture: the foundational principle, its recursive extension, its authority distribution, its access channels, and its conflict-handling.

B4.28 establishes the foundational pair: human governance (A1.01 — the authority-not-labor commitment from Paper 1) and recursive governance (B1.20) together specify that governance is not delegated from the top down but instantiated at every level, with humans retaining the three rights (inspect, modify, override) at each level independently. B4.28 is the most foundational pair in Phase B4, because it grounds the entire recursive governance architecture in Paper 1's governance commitment.

B4.15 establishes how recursive governance distributes authority: the authority distribution commitment (A2.47 from Paper 1) specifies that different human principals hold authority at different levels, and recursive governance (B1.20) is what makes multi-principal authority coherent — each principal's authority is scoped to the level at which they govern.

B4.14 establishes the access channel architecture: cross-level access (B1.19) and recursive governance (B1.20) together specify when higher-level principals can reach below their normal scope. Cross-level access is a governed channel with its own orchestration rules, operating within the recursive governance framework.

B4.26 establishes conflict-handling as the final element of governance authority: conflict as a first-class object (A1.03 from Paper 1) and directed selection (B1.14) together specify how governance handles disagreement. Conflicts are preserved at the substrate level and resolved through directed selection — a governed process in which humans at the appropriate level specify selection criteria that determine which conflicting position is retained.

3. How clusters interact — cross-cluster pairs

The seven clusters are not independent modules. Several pairs serve as cross-cluster connectors, each establishing a structural dependency between two different functional sub-systems. Reading the cross-cluster pairs reveals the architecture's skeleton.

Instinct/reasoning ↔ evolution. B4.18 (B1.13 + B1.01) belongs to Cluster 1 but connects to Cluster 4: mutation governance (B1.13) is one of the three evolution mechanisms, and its relationship to the instinct/reasoning separation (B1.01) is what makes the three mechanisms non-interchangeable. Instinct evolution changes the LLM layer; DNA evolution changes the reasoning layer; action-feedback evolution operates between them.

Lifecycle ↔ evolution. B4.21 (B1.09 + B1.14) and B4.22 (B1.11 + B1.14) belong to Cluster 3 but connect to Cluster 4: directed selection (B1.14) is an evolution mechanism that operates through lifecycle events. Birth and death are the architectural moments at which selection pressure becomes operative. The lifecycle and evolution clusters are not independent sub-systems — they are the temporal and mechanism dimensions of the same adaptive process.

Substrate and record ↔ domain and composition. B4.24 (B1.18 + B1.06) belongs to Cluster 5 but connects to Cluster 6: the DNA layer is what content-domain reads when asking pattern questions. The substrate's record architecture is not separable from the domain and composition architecture; the domain is a reading surface over the substrate's record structure.

Three-level structure ↔ recursive governance authority. B4.05 (B1.02 + B1.20) belongs to Cluster 2 but connects to Cluster 7: the three levels are not just structural arrangements; they are governance scopes. Recursive governance (B1.20) is what gives the three-level structure its authority architecture. Without recursive governance, three levels would be organizational hierarchy; with it, three levels are three independent instantiations of the full CKS governance commitment.

Evolution ↔ recursive governance authority. B4.27 (B1.15 + A1.10) belongs to Cluster 4 and invokes a Paper 1 commitment (A1.10 — the determinism contract) central to the substrate and record cluster. The determinism contract is not just an evidence quality requirement; it is a governance requirement — ungovernable evidence makes directed selection ungovernable, which makes evolution ungovernable.

These cross-cluster connections reveal which commitments are load-bearing across the full architecture: B1.14 (directed selection) appears in Clusters 3, 4, 5, and 7; B1.20 (recursive governance) appears in Clusters 2, 3, and 7; B1.01 (instinct/reasoning) is the root concept that Cluster 1 elaborates and that Clusters 4 and 5 depend on. The cross-cluster connectivity is not accidental — it reflects the genuine structural interdependencies that Paper 2 establishes.

4. The interdependency map as prior art

The 28 composition pairs, organized into seven clusters, constitute the governance interdependency map of Paper 2's three-level instinct/reasoning-separated architecture. The prior art value of this map lies in its specificity and completeness.

Specificity: each pair is named by its component commitments, given a derivation note ID, and characterized with an operational description of the interdependency it formalizes. A pair is not a vague claim that "X and Y interact" but a precise statement that X requires Y for the architectural reason given in the derivation note, derivable from named sections of Paper 2.

Completeness: the map covers 28 distinct interdependencies. Future work that identifies additional composition pairs within Paper 2's commitments will extend the map; it will not challenge the prior art established for the 28 pairs already named. Any architectural claim about a pair that has been named in Phase B4 cannot be presented as a novel discovery — the interdependency was publicly identified, dated, and attributed in the Phase B4 note for that pair.

The seven-cluster organization adds a second layer of prior art. Not only are the 28 individual interdependencies prior art; the higher-order structural grouping is prior art. The claim that Paper 2's governance architecture organizes into seven functional sub-systems — instinct/reasoning substrate, three-level structure, lifecycle governance, evolution architecture, substrate and record, domain and composition, recursive governance authority — is itself a formalized derivation from the source paper, dated and attributed in this note.

This is the defensive purpose of synthesis. Individual pairs protect individual interdependencies. The synthesis protects the architectural map as a whole: the way the architecture decomposes, the way sub-systems relate, and the positions of load-bearing commitments within the network.

5. Phase B4 closure and Phase B5 preview

Phase B4 is now complete. The phase consists of 30 notes: B4.01 (framing note), B4.02 through B4.29 (28 individual pair notes), and B4.30 (this synthesis note).

The 30 notes of Phase B4 are the most structurally complex phase in the Series B derivation program. Phase B1 named the commitments; Phase B2 decomposed them; Phase B3 named what violates them. Phase B4 is what establishes how they depend on each other — the architecture's wiring diagram.

Phase B5 — Operational tests follows Phase B4. Phase B5 will document approximately 15 notes, each isolating one operational test from the Phase B1 foundational commitments or from the composition pairs and making the test a standalone, Zenodo-deposited prior art document. The Phase B5 notes follow the same pattern as A5 notes in Series A: a named test, its full derivation from the source paper, and its complete operational specification as a checklist applicable to any candidate system.

Phase B5 serves a distinct defensive purpose from Phase B4. Composition pairs protect the architecture's wiring. Operational tests protect the architecture's verification surface — the specific conditions under which a system either instantiates or fails to instantiate Paper 2's commitments. A party who implements a three-level instinct/reasoning-separated architecture without meeting the operational tests cannot claim to have implemented the CKS architecture, and cannot claim the operational tests as novel if they have been publicly dated and attributed in Phase B5.

Phase B6 — Boundary cases — will follow Phase B5 with approximately 15 notes covering edge scenarios and boundary conditions: where the three-level structure is ambiguous, where recursive governance encounters conflicting scope claims, where lifecycle events have unclear level attribution, and where evolutionary mechanisms interact in unusual configurations. Boundary case documentation is the final layer of the defensive publication program.

6. Conclusion

The 28 composition pairs of Phase B4, organized into seven higher-order clusters, constitute the governance interdependency map of Paper 2's three-level instinct/reasoning-separated architecture. The map is prior art for the complete network of architectural dependencies: each named pair is an attributed, dated formalization of an interdependency that cannot be claimed as a novel discovery. The seven-cluster organization is prior art for the architectural map as a whole.

Phase B4 is closed. The defensive publication record for Paper 2's architecture now includes the foundational commitment notes of Phase B1, the operational variant and anti-pattern notes of Phases B2 and B3, the 30 composition pair notes of Phase B4, and the cross-derivation notes of Series C. Phase B5 will extend that record to the verification surface.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance.* April 2026. ORCID: 0009-0004-8065-3235.

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